Choosing Expanded or Factored Form
Answer Key

Algebra 1

Quick Answers

1. $x^2 + 5x - 36$
2. $(x+3)(x+7)$
3. 0, 4 s
4. $3x^2 - 15x + 2$
5. $2(x-3)(x+3)$
6. (a) factored form = $(x-2)(x-4)$, (b) the zeros = 2, 4
7. $4x^2 - 12x + 9$
8. $x(x-2)(x+2)$
9. (a) constant term = 15, —
10. $(2x+3)(3x-4)$
11. (a) simplified = $2x^2 + 6x$, (b) factored form = $2x(x+3)$
12. (a) factored form = $(x+2)(x+5)$, (b) area at x = 3 = 40 m^2, —

What is verified

10 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSA-APR.A.1 – *problems 1, 4, 7, 9, 11, 12*
HSA-APR.B.3 – *problems 3, 6*
HSA-SSE.B.3 – *problems 2, 5, 6, 8, 10, 11, 12*
Difficulty 1–5 of 5 across 17 tagged checks.

1. Write $(x-4)(x+9)$ in standard form.
Solution: Four products, then collect the middle pair: $x^2+9x-4x-36 = x^2+5x-36$. The constant term -36 is the product of the two constants, and it is invisible until the brackets come off. $\boxed{x^2 + 5x - 36}$

2. Write $x^2 + 10x + 21$ in factored form.
Solution: Two numbers with product 21 and sum 10: since $21 = 3\times7$ and $3+7 = 10$, the factors are $(x+3)$ and $(x+7)$. $\boxed{(x+3)(x+7)}$

3. $h(t) = -5t(t-4)$ metres: find both times at ground level.

Solution: The height is already factored, and a product is zero exactly when one of its factors is zero. So $-5t = 0$ gives $t = 0$, and $t - 4 = 0$ gives $t = 4$. No expanding and no quadratic formula is needed — that is the whole advantage of the factored form here. The ball leaves the ground at $t = 0$ and lands at $t = 4$. $\boxed{t = 0 \text{ s} \quad \text{or} \quad t = 4 \text{ s}}$

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4. Write $3x(x - 5) + 2$ in standard form.

Solution: Distribute first: $3x \cdot x = 3x^2$ and $3x \cdot (-5) = -15x$. The $+2$ is outside the product and simply follows: $3x^2 - 15x + 2$. $\boxed{3x^2 - 15x + 2}$

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5. Write $2x^2 - 18$ in fully factored form.

Solution: Common factor first: $2x^2 - 18 = 2(x^2 - 9)$. What is left is a difference of two squares, $x^2 - 3^2$, which factors as $(x - 3)(x + 3)$. $\boxed{2(x - 3)(x + 3)}$

Skipping the common factor and writing $(2x-6)(x+3)$ is not fully factored; the first bracket still has a 2 in it.

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6. $g(x) = x^2 - 6x + 8$: (a) rewrite to show the zeros; (b) give the zeros.

Solution (a): Two numbers with product 8 and sum -6 are -2 and -4.

$\boxed{(x - 2)(x - 4)}$

Solution (b): A product is zero exactly when a factor is zero, so $x - 2 = 0$ or $x - 4 = 0$.

$\boxed{x = 2 \quad \text{or} \quad x = 4}$

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7. Write $(2x - 3)^2$ in standard form.

Solution: A squared binomial is the bracket times itself, so it has three terms: $(2x)^2 = 4x^2$, twice the cross product $2 \cdot 2x \cdot (-3) = -12x$, and $(-3)^2 = 9$. $\boxed{4x^2 - 12x + 9}$

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8. Write $x^3 - 4x$ in fully factored form.

Solution: Every term has an x, so take it out first: $x^3 - 4x = x(x^2 - 4)$. The bracket is a difference of two squares: $x^2 - 2^2 = (x - 2)(x + 2)$. $\boxed{x(x - 2)(x + 2)}$

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9. $P(x) = (x + 5)(x - 1)(x - 3)$: (a) the constant term; (b) which form and why.

Solution (a): The constant term of a polynomial is its value at $x = 0$, because every other term carries at least one factor of x. Substituting into the factored form: $P(0) = (0 + 5)(0 - 1)(0 - 3) = (5)(-1)(-3)$. $\boxed{15}$

Solution (b) — instructor-judged. The factored form did the work. Substituting $x = 0$ into three short brackets is three easy products; reading the constant term out of standard form would have meant expanding the whole cubic first. Full credit compares the effort the two forms demand; repeating the value 15 is not an explanation.

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10. Write $6x^2 + x - 12$ in factored form.

Solution: With a leading coefficient of 6, split the middle term. Two numbers with product $6 \times (-12) = -72$ and sum 1 are 9 and -8:

$$\begin{aligned} 6x^2 + x - 12 &= 6x^2 + 9x - 8x - 12 \\ &= 3x(2x+3) - 4(2x+3) \;=\; (2x+3)(3x-4). \end{aligned}$$

$\boxed{(2x+3)(3x-4)}$

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11. $(x+3)(x-3) + (x+3)^2$: (a) expand and simplify; (b) factor the result.

Solution (a): $(x+3)(x-3) = x^2 - 9$ and $(x+3)^2 = x^2 + 6x + 9$. Adding them, the -9 and $+9$ cancel: $2x^2 + 6x$. $\boxed{2x^2 + 6x}$

Solution (b): Both terms share $2x$: $2x^2 + 6x = 2x(x+3)$. Notice the shortcut the factored view offers — $(x+3)$ was a factor of both starting pieces all along. $\boxed{2x(x+3)}$

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12. Garden of area $A(x) = x^2 + 7x + 10$ square metres: (a) factor; (b) area at $x = 3$; (c) which form gives the sides?

Solution (a): Two numbers with product 10 and sum 7 are 2 and 5.

$\boxed{(x+2)(x+5)}$

Solution (b): Substituting $x = 3$ into either form gives the same number. Using standard form: $9 + 21 + 10 = 40$. Using the factored form as a check: $(3+2)(3+5) = 5 \times 8 = 40$.

$\boxed{40 \text{ m}^2}$

Solution (c) — instructor-judged. The factored form. A rectangle's area is the product of its two sides, and $(x+2)(x+5)$ is exactly that product written down, so the sides are $(x+2)$ m and $(x+5)$ m — at $x = 3$, 5 m by 8 m. Standard form gives no sides at all until it is factored. Full credit ties the two factors to the two side lengths; giving the numbers without that reason earns half credit.